



AKGEC SKILLS FOUNDATION

Training Program Details

CENTRE NAME- BOSCH AUTOMOTIVE CENTRE OF EXCELLANCE

Training/Course Title:	Winter Training Program on Automotive Technology
Target Audience:	B.Tech Students
Fee:	11000/-
Duration:	80 Hrs.
Module	• Introduction to automobiles covering history, evolution, vehicle classification, and complete vehicle architecture with an understanding of major subsystems. • Study of chassis structures, suspension systems, axle types, and their role in vehicle stability, comfort, and load distribution. • Understanding steering geometry parameters such as camber, caster, toe, KPI, along with steering mechanisms, wheel types, and tyre construction and maintenance. • Detailed learning of transmission systems including clutch operation, manual and automatic gearboxes, differential, propeller shaft, and complete driveline behavior. • Comprehensive knowledge of internal combustion engines including types, working cycles, engine components, dismantling, assembly, cooling systems, and lubrication systems. • Fuel delivery systems including fuel tank, pump, filter, injectors, MPFI operations, carburetion basics, and diagnostics of petrol fuel systems. • Automotive electrical systems covering battery, starter motor, alternator, vehicle wiring harness, circuits, and testing using electrical benches and tools. • Engine management system comprising ECU architecture, sensors, actuators, electronic control strategies, fault codes, and diagnostics using KTS or similar scan tools. • Braking systems including hydraulic brakes, brake components, disc and drum mechanisms, ABS working principles, and demonstration of modern safety systems like ESP, EBS, and airbags. • HVAC and workshop practical skills including automotive AC system operation, service equipment usage, leak

	detection, wheel alignment procedures, vehicle inspection practices, and final assessment activities.	
Outcome:	• Ability to understand and explain the structure, functioning, and integration of all major automobile systems. • Capability to dismantle, inspect, assemble, and test key components such as engines, gearboxes, clutches, brakes, and AC systems. • Competence in reading and interpreting wiring diagrams and performing electrical system testing for battery, starter, alternator, and vehicle circuits. • Skill to diagnose engine and electrical faults using tools like KTS, interpret fault codes, and perform effective troubleshooting. • Practical proficiency in vehicle maintenance operations including wheel alignment, AC servicing, system testing, and understanding safety technologies. • Readiness for workshop and diagnostic roles with confidence in performing real-world vehicle inspection, servicing, and repair tasks.	
Certification:	The participants, on successful completion of above trainings, will receive joint certification by BOSCH India & AKGEC.	
Program Coordinator:	Name: Mr. Vikram Singh Rajput	Contact number: 8851805766
Payment Mode:		